Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Williams, Forman A. Ph.D.		
Project Title:	Droplet Combustion Experiment (DCE/DCE-2)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	858 534-5492
Organization Name:	University of California, San Diego		
PI Address 1:	Center for Energy Research		
PI Address 2:	Department of Mechanical and Aerospace Engineering		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	12/14/1999	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	This is a collaborative effort of NASA Glenn Research Center, the University of California, San Diego and Princeton University, to advance scientific knowledge of droplet combustion. Experiments are planned for space platforms, including the International Space Station. By reducing buoyancy effects, space-based experiments contribute to improved understanding of droplet combustion. In preparing for space experiments, ground-based microgravity experiments and theoretical and computational analyses are performed by the co-investigators, and relevant instrumentation is improved. The objective of this collaboration is to provide scientific support for fundamental, single component, isolated (free and tethered) droplet burning experiments to be performed on board space platform. The specific objectives of these studies are to provide fundamental results which impact fire safety issues in low gravity environments and to advance the		

Task Description:	scientific understanding of droplet burning/flame/extinction dynamics over characteristic times which can not be studied in ground based configurations. Engineering and design support functions contribute to: developing and refining methods to perform isolated droplet combustion experiments in general; determining and testing methods to acquire appropriate experimental observations; developing and evaluating methods for presenting and analyzing the collected data. Experimental support functions include: theoretical analyses to define appropriate experimental conditions and to analyze the experimental results obtained; bench scale and drop tower e xperiments to test and verify methodologies to be applied in the space-based experiments. The overall purpose of the research is to achieve fundamental advances in the science of droplet combustion, which is central to a number of energy-producing devices and systems. In particular, unsteady liquid and gas phenomenon, and extinction chemistry of normal alkanes and simple alcohol's are investigated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 03/07/2024)